AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous ML discovery through pure evidence.

PERSISTENT WORLD

FRESH SCIENTIST

EVIDENCE FIRST

Problem → Motivation → System → Experience → Evaluation → Insights

01 / PROBLEM · THE RESEARCH TASK

Can an agent carry a research task to completion?

Turn a task contract into a tested model, a reproducible implementation, and evidence another researcher can inspect.

Explore	Choose hypotheses, write code, and run experiments.	Scientific judgment
Recover	Handle failures and revise unsuccessful ideas.	Long-running work
Deliver	Retain a valid model and the evidence behind it.	Verifiable progress

Our test case: short-video ranking on KuaiRand-Pure, with a fixed evaluator and public-validation boundary.

01 / PROBLEM · FAILURE MODES

Why Current Research Agents Fall Short

Three obstacles to sustained autonomous research.

01 CLOSED-WORLD WORKFLOW

Observe → Hypothesize → Code → Evaluate

ERROR

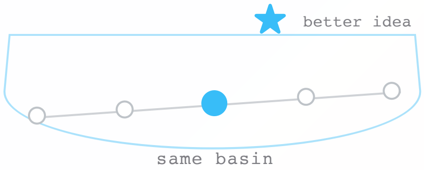
↓

dead end

Predefined tools and recovery paths cannot handle the unexpected.

Unexpected problems require developer intervention.

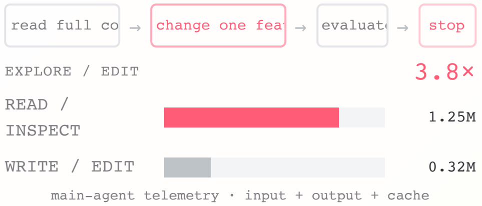
02 SINGLE TRAJECTORY



A single-agent search inherits its previous pattern and its unchallenged assumptions.

It can narrow exploration and carry errors forward.

03 TREE SEARCH



Each child makes one narrow change, then exits. The next child pays the context cost again.

One small patch. A full re-read.

DESIGN RESPONSE

Research is open-world, non-linear, and exploratory.

ADAPT DIVERSIFY LEARN

01 / PROBLEM · SCIENTIFIC VALIDITY

A single agent can be wrong while the run looks healthy

The same context can choose the test, run it, and explain the result. **That makes errors easy to carry forward.**

ONE CONTEXT OWNS THE LOOP

DESIGN the claim → IMPLEMENT the test → MEASURE the metric → INTERPRET the result

A mistake at the first step can survive the rest of the trajectory.

GEMINI · CLAIM / IMPLEMENTATION

CLAIM	within-user BPR
ACTUAL	cross-user pairing

The code does not test the mechanism it names.

GEMINI · FEATURE CONSTRUCTION

CLAIM	dense target statistics
ACTUAL	self / future-label leakage

A plausible gain is not automatically leak-free evidence.

Research is an open-world task.
We need an open-world coding agent.

01 OPEN ACTION SPACE

- Shell
- Files
- Browser
- Git

Use what exists

02 SELF-RECOVERY

- Read errors
- Inspect docs
- Inspect source
- Install packages

Fix what breaks

03 RUNTIME EXTENSION

- MCP
- Skills
- CLI
- Packages

Add what's missing

WORKFLOW EXAMPLE · LANGGRAPH

LangGraph makes a known workflow explicit as nodes + edges.

Missing capability? Leave the graph → inspect → extend → retry

Preserve the research world.

Reset the researcher.

Pass evidence, failures, and code to a fresh Scientist.

Two scientific roles. One safe runtime.

Scientist	Hypothesize · code · test · interpret	One trajectory
META	Audit · preserve · hand off	Across trajectories
Runtime	Isolation · cancellation · invariants	System lifetime

The Scientist chooses the science. META maintains the world. Runtime enforces mechanics, not scientific judgment.

03 / SYSTEM · PERSISTENCE BOUNDARY

The persistence boundary is explicit.

Environment

- Task contract
- Curated data + evaluator
- Starter baseline
- Budget + permissions

→

Research World

A · VERIFIED / EVIDENCE

- Verified knowledge
- Evidence ledger

B · CURATED / FALLIBLE

- Research brief
- Fallible priors

C · IMPLEMENTATION

- Implementation State

What persists across trajectories

CONTEXT RESET

No previous reasoning trajectory

Fresh Scientist

Scientist

owns scientific judgment

- reads the research world
- forms hypotheses
- runs experiments
- interprets evidence
- writes a report

inherits experience, not momentum

→

META Review

META

owns what survives

- audits claim ↔ evidence validity
- scopes conclusions
- maintains shared research memory
- crystallizes retained implementation into State

Selective persistence / trajectory handoff

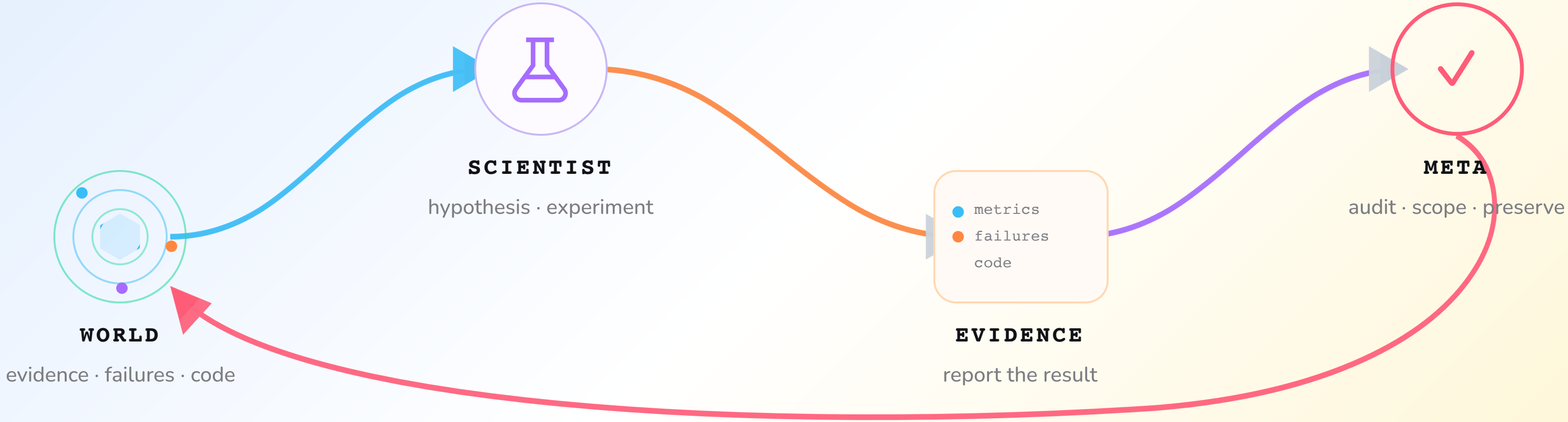
audit → scope → preserve

Runtime — deterministic mechanics

- Workspace isolation
- Runner + evaluator
- Logging
- Artifact capture
- State operations

03 / SYSTEM · THE HANDOFF LOOP

Each cycle leaves a world to build on.



SELECTIVE PERSISTENCE

updated world → fresh Scientist

PERSISTS metrics · code · failures · original reports

RESTARTS an independent reasoning trajectory

03 / SYSTEM · RESEARCH MEMORY

Keep facts distinct from intuition.

```
research world
├── evidence & provenance
├── verified knowledge
├── current research state
├── fallible priors
└── State + implementation
```

Built to be re-examined.

Facts have evidence.

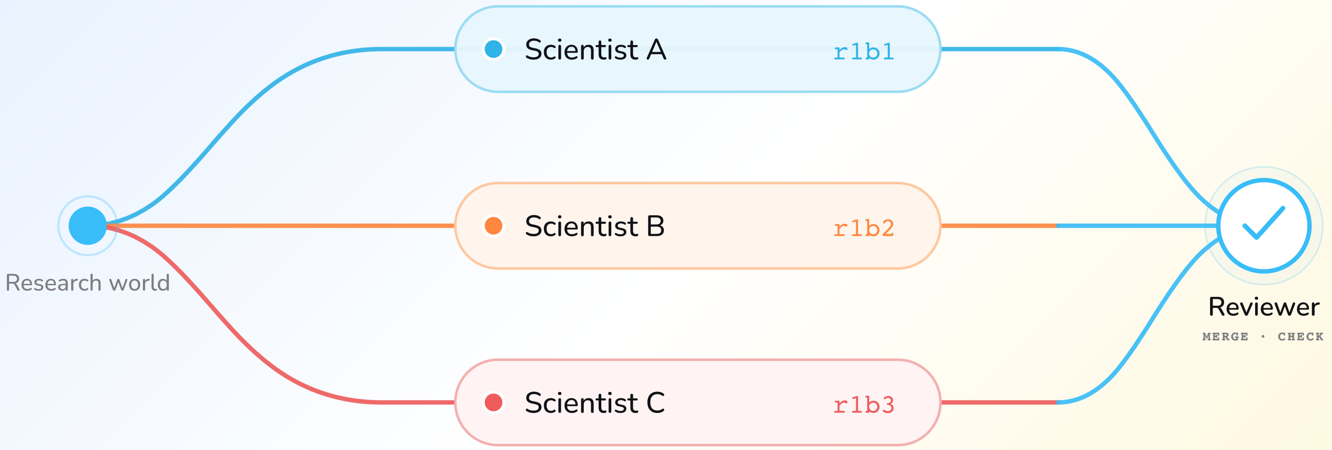
Priors can be overturned.

Code can be picked up again.

Original reports remain available for audit.

03 / SYSTEM · PARALLEL EXPLORATION

One starting point. Independent answers.



Explore in isolated worktrees. Review after completion.

Optional synthesis: one implementation parent + reference evidence from other branches.
Adoption is explicit via parallel-promote. Code and memory are not merged automatically.

Start with the task and working constraints.

The operator defines the objective and constraints; the agent chooses and executes the experiments.

01 `task.md`

What to solve.

Objectives · constraints · evaluation

EXAMPLE

Rank short videos for each user and interaction context.

KuaiRand-Pure · long_view · GAUC + nDCG@5

02 `PERSONAL.md`

How you want it solved.

Preferences · research style · priorities

EXAMPLE

Run on macOS / Apple Silicon with uv.

≤ 15 min / experiment · never print credentials

Autonomous by default. Supervisable by design.

Evidence changed the next experiment.

E001–E013: reject weak ideas, improve representations, then test ensembles.

E001	Screen historical target encodings; reject weak variants.	Medium screening
E003	Establish a valid rich-FM checkpoint.	Full · 0.6016310
E008	Retain a five-seed, 38-field FM ensemble.	Full · 0.6040901
E013	Select the eight-seed, 46-field FM ensemble.	Full · 0.6059363

These are milestones within four autonomous cycles. Medium screens ideas; Full evaluations support retained score claims.

04 / EXPERIENCE · RECOVERY IN THE RUN

An evaluation failure became a recoverable step.

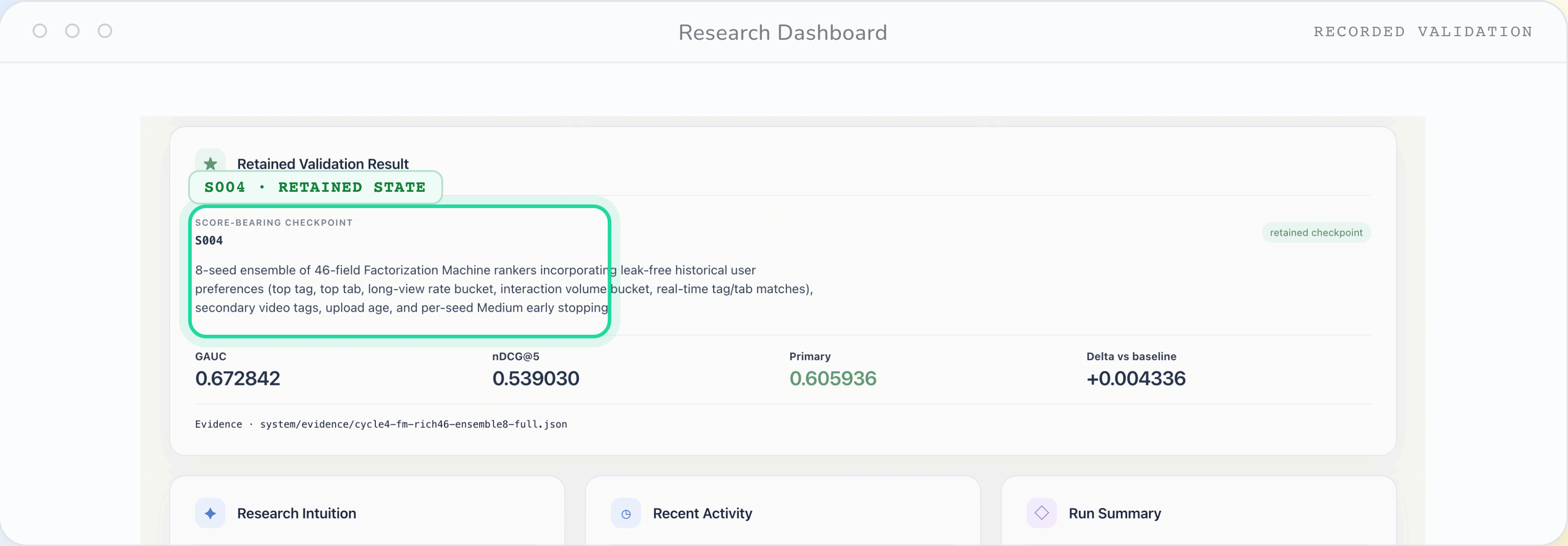
Cycle 1 completed training and evaluation, then failed while writing the evidence file.

Failure	NumPy float32 values broke JSON serialization.	Evidence writer
Repair	The Scientist converted scalars and reran.	Agent-authored fix
Retain	Printed measurements and saved evidence survived.	Auditable recovery

No human selected this repair or edited the code. This is one observed recovery case, not a guarantee for every failure.

04 / EXPERIENCE · INSPECT THE RESULT

A retained state keeps the evidence inspectable.



Inspect the retained state and the evidence behind it.

Retained validation checkpoint · evidence path · reproducible implementation.

05 / EVALUATION · PUBLIC-VALIDATION RESULT

Autonomous research. Measured gains.

PRIMARY / RESEARCH AGENT

0.6059363

+0.0043363 absolute improvement

METRIC	Official FM	Ours
GAUC	0.6674000	0.6728421
nDCG@5	0.5357000	0.5390304
Primary	0.6016000	0.6059363

Primary = mean(GAUC, nDCG@5). Public validation only; hidden-test scoring is organizer-controlled.

Final recipe: 46 categorical features × 8-seed FM ensemble · NumPy / CPU

An audit trail from experiment to output.

4

Autonomous cycles

Within a 50-iteration cap

13

Named experiments

E001–E013 · 7 Full evaluations

0

Manual scientific interventions

After launch · 0 GPU-hours

170,588 prediction rows passed the unchanged Starter Kit alignment checker.

LLM tokens: 48.24M including cache reads; 4.02M excluding cache reads.

Resource use is part of the evidence.

Project-level subscription estimate, with run-level telemetry reported separately.

~$10

from first line of code to final result

coding · debugging · agent runs · all experiments

MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire research loop.

COMPUTE

1 × MacBook M2

All development and experiments ran on a single consumer laptop.

The recorded runs used CPU training; LLM usage remains a separate cost.

~$10 is the estimated subscription-equivalent usage across the entire project, including coding, debugging and all experiments — not cost per experiment.

Compare scores alongside their evidence.

Runs differ in model, budget, and evidence quality; this is not a controlled causal comparison.

The run taught us more than the final score.

What survived the experiments — and what remains unsettled.

01

Diversity is a prior.

Keep `gemini-3.7-flash` as META while Scientist rotates across `gpt-5.6-sol`, `gemini-3.7-flash`, and `gpt-5.6-luna`. This widens priors; it does not prove the gain.

02

A score is not a claim.

Audits found leakage and sampling mismatches in competing experiments. META can scope evidence; it is not a formal verifier.

03

Failure can be evidence.

After evaluation, NumPy `float32` values broke serialization. The Scientist fixed the writer, reran, and kept the measurements.

04

Reset ≠ no anchor.

Fresh cognition reopens the search, but shared summaries can still omit context or anchor future work. Share evidence later than cognition.

OBSERVED, NOT PROVEN

Parallel/Synthesis reached public-validation Primary 0.6055536; the canonical E001–E013 frontier remains 0.6059363.

4 cycles

0 post-launch interventions

0 GPU-hours

06 / INSIGHTS · LIMITS AND NEXT EXPERIMENTS

Reproducible evidence. Open questions.

01	One task does not establish generality.	External validity
02	META, State, and reset are not isolated.	Causal attribution
03	Summaries can lose context and anchor.	Memory quality

Next questions: broader tasks, targeted ablations, and less repeated execution.

A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

01 MANY AGENTS

Give every agent a branch.

main

clean trunk

- agent A / branch
- agent B / branch
- agent C / branch

review

→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

02 NO QUOTA / NO LAPTOP

Leave the next move in GitHub.

01 Issue / prompt — inject the task → 02 GitHub Action — run the agent → 03 Commit — review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

PROPOSED PRACTICE Workflow guidance; GitHub continuation is not a benchmarked system feature.

Keep the experience.
Reopen the search.

```
# From the configured repo, run one autonomous cycle
$ ./scripts/research-agent step --cli codex \
    --target /absolute/path/to/project --allow-edits
```

github.com/zc6600/research-agent-kuairand ↗

Code · Technical report · Experiment ledger · Checked output

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD

CZ

Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture · integration

ZZ

Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles · evidence review

SX

Shilin Xu

DATA / METRICS

Data workflow · contracts · alignment

GG

GE GAO

RUNTIME / RELIABILITY

Execution · testing · integration support

JL

Jiran Li

EVIDENCE / REPRODUCIBILITY

Telemetry · documentation · packaging

Every handoff kept the research world moving.

Research is an open-world task.
We need an open-world coding agent.

LANGGRAPH-STYLE ORCHESTRATION

predefined graph

fixed tool set

Tool A

Tool B

Tool C

API fails

↓

no recovery tool

↓

DEAD END

Capabilities are bounded by what developers anticipated.

CODING-AGENT HARNESS

open action space

Computer

↓

shell | files | browser

↓

git | docs | packages

skills → agent ← MCP

↓

CLI / API

fail → inspect → learn → modify environment → retry

The agent can acquire the missing capability at runtime.